Aaryan Lath

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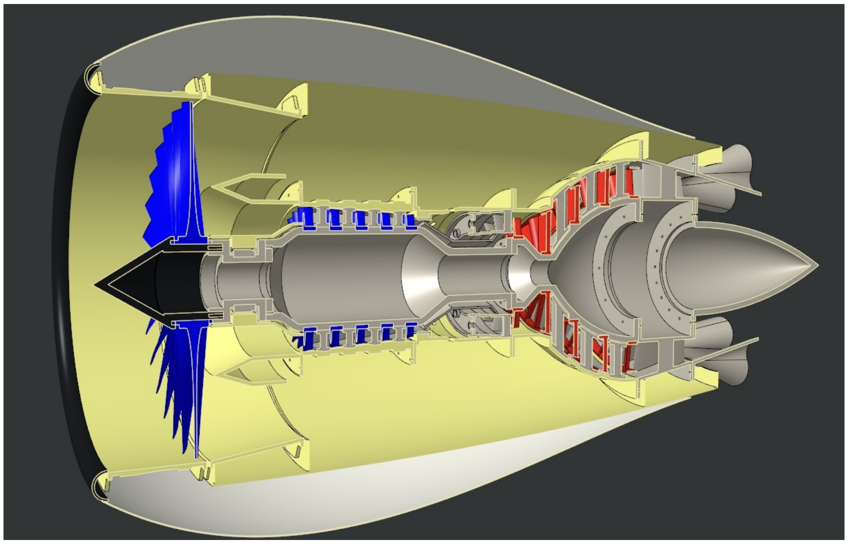
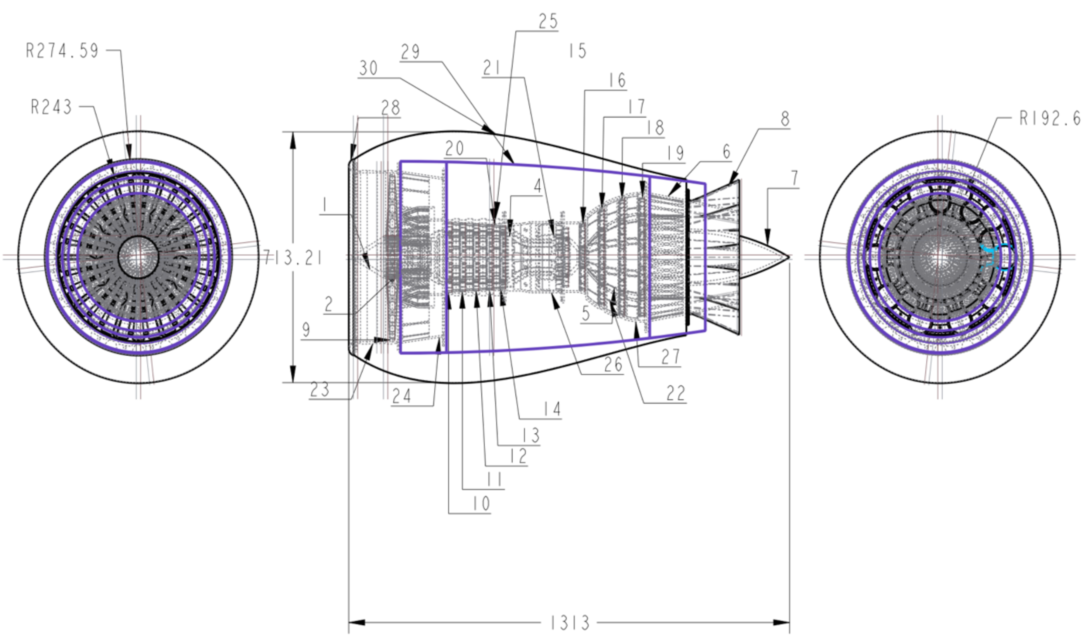
Jet-Engine



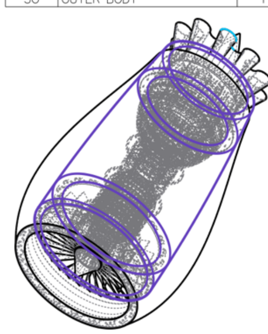
Top-Down assembly design of a turbofan engine for understanding individual components.

How:

- Employed **CREO Parametric** for the model
- Applied **GD&T** on all components
- Completed a complete model with **80 hours** of modelling.



ITEM	PC. NO.	QTY.
1	NOSE-CONE	1
2	PRIMARY-TURBINE-BUSHING	1
3	BEARING-ONE	1
4	CENTRAL-MOUNTING-SHAFT	1
5	REAR-MOUNTING-SHAFT	1
6	REAR-CASE	1
7	END-CONE	1
8	EXHAUST-CASE	1
9	PRIMARY-TURBINE-BLADES	1
10	AIR-TURBINE-ONE	1
11	AIR-TURBINE-TWO	1
12	AIR-TURBINE-THREE	1
13	AIR-TURBINE-FOUR	1
14	AIR-TURBINE-FIVE	1
15	FUEL-TURBINE-ONE	1
16	FUEL-TURBINE-TWO	1
17	FUEL-TURBINE-THREE	1
18	FUEL-TURBINE-FOUR	1
19	FUEL-TURBINE-FIVE	1
20	AIR-TURBINE-BRACKET	1
21	FUEL-INTAKE	1
22	FUEL-TURBINE-BRACKET	1
23	PRIMARY-TURBINE-CASE	1
24	BEARING-CASE	1
25	AIR-TURBINE-CASE	1
26	FUEL-INTAKE-CASE	1
27	FUEL-TURBINE-CASE	1
28	FRONT-RIM	1
29	INNER-BODY	1
30	OUTER-BODY	1



Tolerances: .XX ± .030 .XXX ± .015 Angular ± 2° Units are in inches.	Drawn By: Aaryan Lath	Turbofan Jet Engine	BYPASS-ENGINE	00
			7/12/25	Done on CREO

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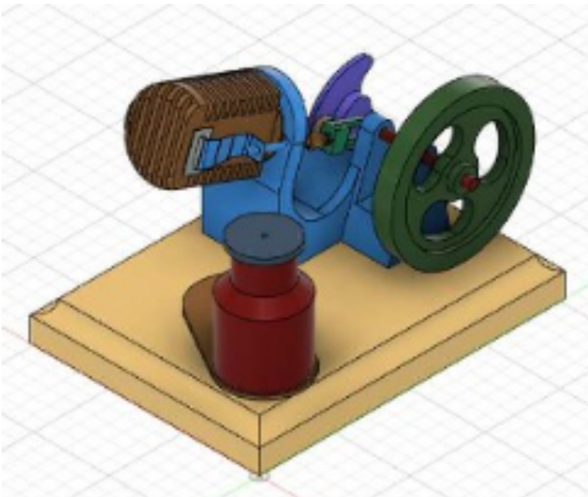
Single-Piston Sterling Engine

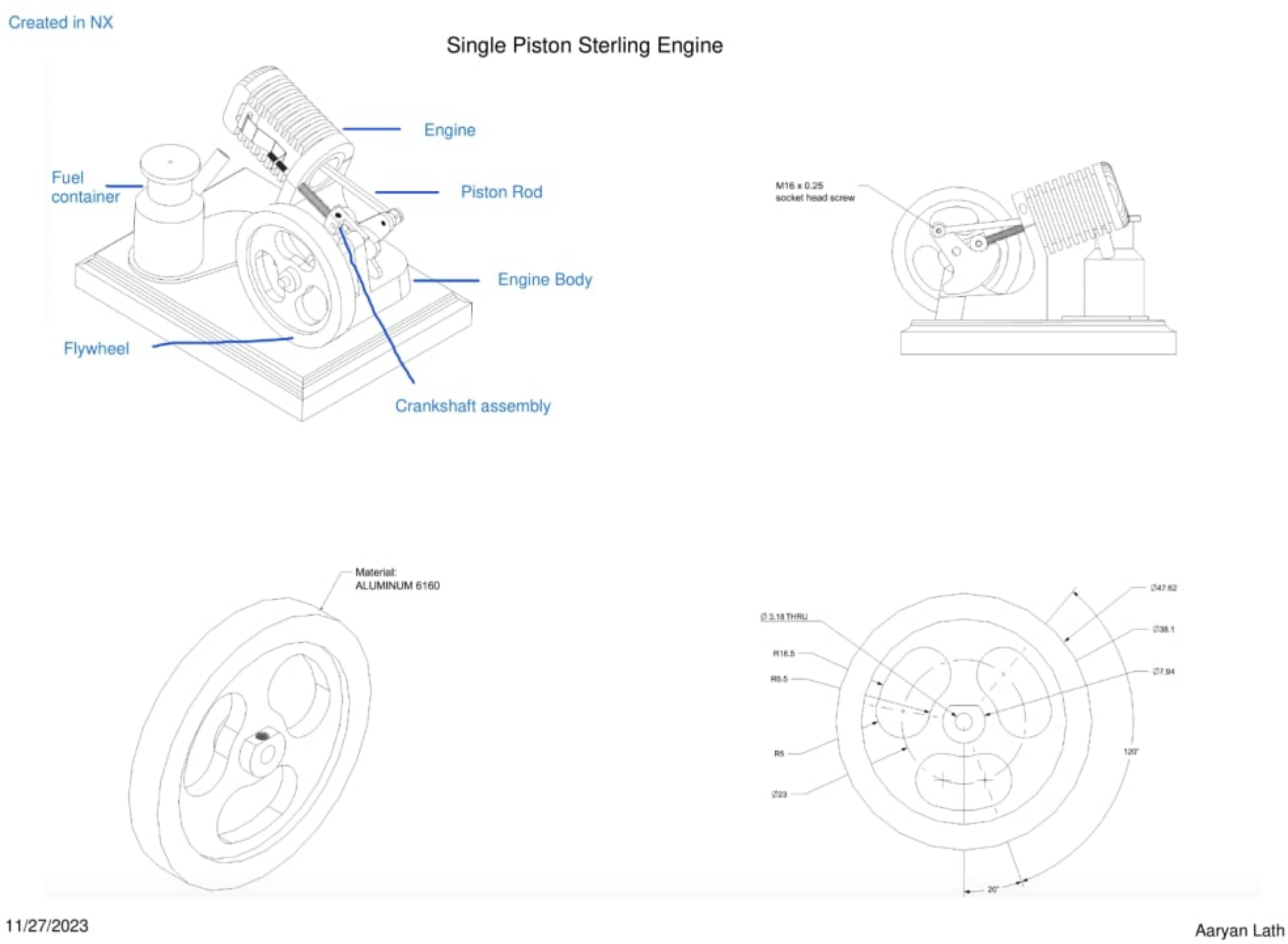


A bottom-up assembly of a Single Piston Sterling Engine as part of my final project in my Graphical Communication and Spatial Analysis course.

How:

- Employed **Siemens NX** for modelling and **ARAS Innovator** for **PDM** and **PLM**
- Applied **GD&T** on all components

Received **recognition** from Professor Travis Fuerst at **Purdue University** and worked with him for deploying **Teamcenter** the following year.



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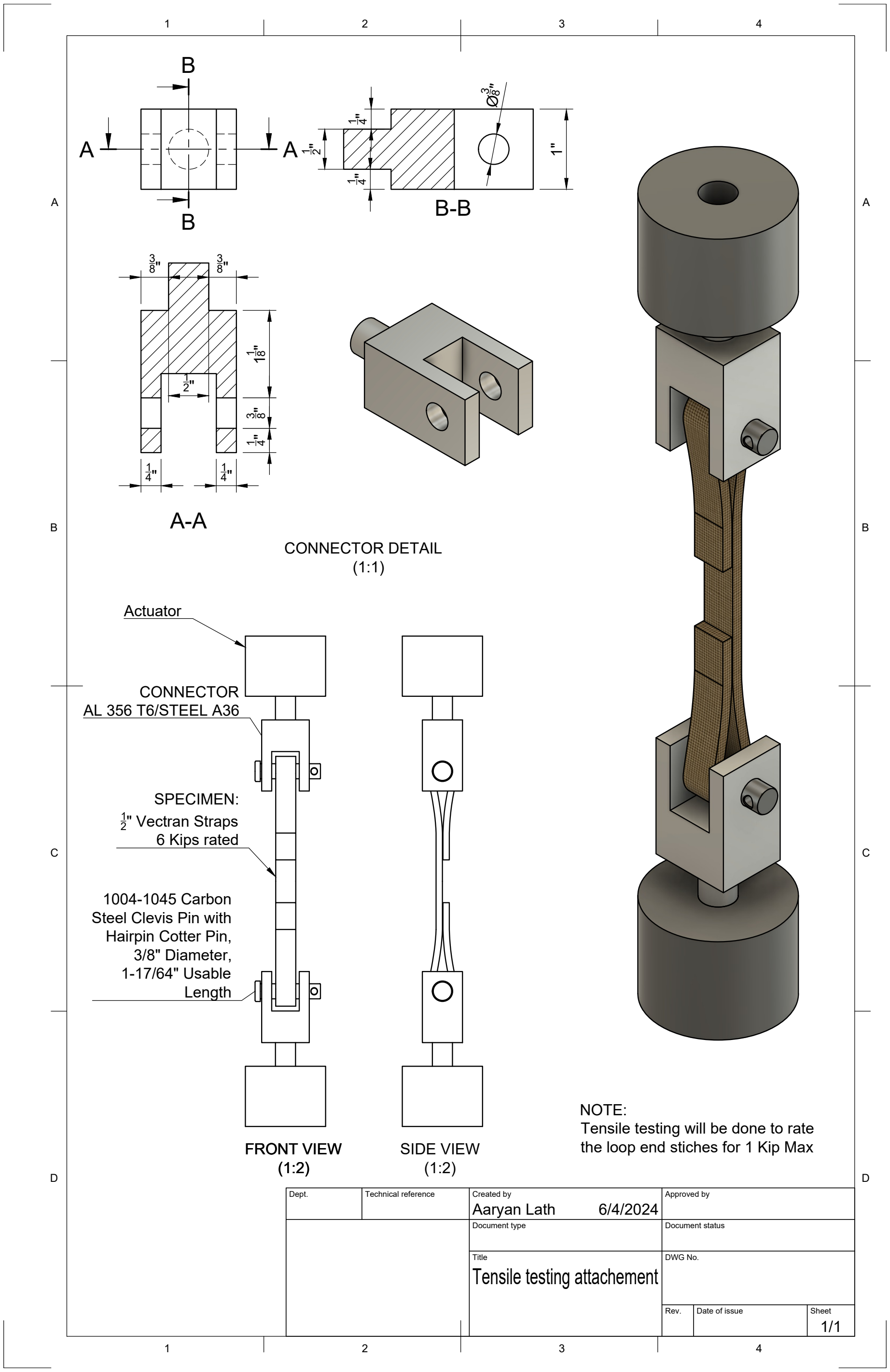
Tensile-test



Designed and modelled an experiment for Tensile testing Vectran straps for Resilient Extraterrestrial Habitat Institute.

How:

- Employed **Fusion 360** for modelling.

- Applied **GD&T** on all components.

Results:

- Hand-stitched the Vectran straps with different stitching patterns and **performed** a tensile test until failure.

- Conducted a **fatigue test** and developed a **rainflow counting algorithm** simulating lunar vibrations.

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3D-Printing Milestones course

Designed and modelled a lightsaber as the final project for the 8 week **Fusion 360 x 3D-printing** course

How:

- Built the lightsaber using only **4** features
- 3D-printed using the **Prusa MK3 PLA 3D** printer.
- Obtained a **Certificate** for “CAD / 3D Printing Milestone Workshop.”